





1 H Hydrogen 1.008																	2 He Helium 4.0026 2																		
Atomic Number → ← Symbol ← Name ← Atomic Weight Electrons per shell →																																			
3 Li Lithium 6.94 1	4 Be Beryllium 9.012 2	State of matter (color of name) GAS LIQUID SOLID UNKNOWN Subcategory in the metal-metalloid-nonmetal border (color of background) Alkali metals Lanthanides Metalloids Unknown chemical properties Alkaline earth metals Actinides Reactive nonmetals Transition metals Post-transition metals Noble gases																5 B Boron 10.81 3	6 C Carbon 12.011 4	7 N Nitrogen 14.007 5	8 O Oxygen 15.999 6	9 F Fluorine 18.998 7	10 Ne Neon 20.180 8												
11 Na Sodium 22.990 3	12 Mg Magnesium 24.305 2	13 Al Aluminum 26.982 3	14 Si Silicon 28.086 4	15 P Phosphorus 30.974 5	16 S Sulfur 32.06 6	17 Cl Chlorine 35.45 7	18 Ar Argon 39.948 8	19 K Potassium 39.098 4	20 Ca Calcium 40.078 2	21 Sc Scandium 44.956 3	22 Ti Titanium 47.88 4	23 V Vanadium 50.942 5	24 Cr Chromium 51.996 6	25 Mn Manganese 54.938 7	26 Fe Iron 55.845 8	27 Co Cobalt 58.933 9	28 Ni Nickel 58.69 10	29 Cu Copper 63.546 11	30 Zn Zinc 65.38 12	31 Ga Gallium 69.723 3	32 Ge Germanium 72.63 4	33 As Arsenic 74.922 5	34 Se Selenium 78.96 6	35 Br Bromine 79.904 7	36 Kr Krypton 83.798 8										
37 Rb Rubidium 85.468 5	38 Sr Strontium 87.62 2	39 Y Yttrium 88.906 3	40 Zr Zirconium 91.224 4	41 Nb Niobium 92.906 5	42 Mo Molybdenum 95.94 6	43 Tc Technetium 98 7	44 Ru Ruthenium 101.07 8	45 Rh Rhodium 102.91 9	46 Pd Palladium 106.42 10	47 Ag Silver 107.87 11	48 Cd Cadmium 112.41 12	49 In Indium 114.82 3	50 Sn Tin 118.71 4	51 Sb Antimony 121.76 5	52 Te Tellurium 127.6 6	53 I Iodine 126.91 7	54 Xe Xenon 131.29 8	55 Cs Cesium 132.91 6	56 Ba Barium 137.33 2	57-71 Lanthanides	72 Hf Hafnium 178.49 6	73 Ta Tantalum 180.948 7	74 W Tungsten 183.84 8	75 Re Rhenium 186.21 9	76 Os Osmium 190.23 10	77 Ir Iridium 192.22 11	78 Pt Platinum 195.08 12	79 Au Gold 196.967 13	80 Hg Mercury 200.59 14	81 Tl Thallium 204.38 3	82 Pb Lead 207.2 4	83 Bi Bismuth 208.98 5	84 Po Polonium 209 6	85 At Astatine 210 7	86 Rn Radon 222 8
87 Fr Francium 223 7	88 Ra Radium 226 2	89-103 Actinides	104 Rf Rutherfordium 261 6	105 Db Dubnium 262 7	106 Sg Seaborgium 266 8	107 Bh Bohrium 264 9	108 Hs Hassium 277 10	109 Mt Meitnerium 268 11	110 Ds Darmstadtium 271 12	111 Rg Roentgenium 272 13	112 Cn Copernicium 285 14	113 Nh Nihonium 284 15	114 Fl Flerovium 289 16	115 Mc Moscovium 288 17	116 Lv Livermorium 293 18	117 Ts Tennessine 289 19	118 Og Oganesson 294 20																		

57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium 145	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.967
89 Ac Actinium 227	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium 237	94 Pu Plutonium 244	95 Am Americium 243	96 Cm Curium 247	97 Bk Berkelium 247	98 Cf Californium 251	99 Es Einsteinium 252	100 Fm Fermium 257	101 Md Mendelevium 258	102 No Nobelium 259	103 Lr Lawrencium 260

Элементы

- ★ С этого момента химики стали искать и находить новые химические элементы
- ★ постепенно заполняя всю периодическую таблицу Менделеева (118 элементов)

The image displays a comprehensive periodic table of elements, color-coded by groups and properties. The table is organized into blocks for s, p, d, and f orbitals, with a legend explaining the color coding and group labels.

Legend:

- Atomic Number:** 1 (Hydrogen)
- Symbol:** H
- Name:** Hydrogen
- Atomic Weight:** 1.008
- Electrons per shell:** 1
- State of matter (color of name):** GAS (red), LIQUID (blue), SOLID (green), PLASMA (yellow)
- Subcategory in the metal-metalloid-nonmetal trend (color of background):**
 - Alkali metals (red)
 - Alkaline earth metals (orange)
 - Transition metals (blue)
 - Lanthanides (light blue)
 - Actinides (light green)
 - Post-transition metals (light yellow)
 - Metalloids (yellow)
 - Reactive nonmetals (light green)
 - Noble gases (pink)
- Unknown chemical properties:** (grey background)

Table Structure:

- Groups (Columns):** 1 (IA), 2 (IIA), 3 (IIIB), 4 (IVB), 5 (VB), 6 (VIB), 7 (VIIB), 8 (VIII), 9 (VIII), 10 (VIII), 11 (IB), 12 (IIB), 13 (IIIA), 14 (IVA), 15 (VA), 16 (VIA), 17 (VIIA), 18 (VIIIA).
- Periods (Rows):** 1 to 7.
- Block Structure:** s-block (groups 1-2), p-block (groups 13-18), d-block (groups 3-10), f-block (lanthanides and actinides).

Elements shown (by group):

- Group 1 (IA):** H, Li, Na, K, Rb, Cs, Fr
- Group 2 (IIA):** Be, Mg, Ca, Sr, Ba, Ra
- Group 3 (IIIB):** Sc, Y, La, Ce, Pr, Nd, Pm, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu
- Group 4 (IVB):** Ti, Zr, Hf, Ta, Nb, Mo, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn
- Group 5 (VB):** V, Nb, Ta, Hf, Ta, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn
- Group 6 (VIB):** Cr, Mo, W, Re, Os, Ir, Pt, Au, Hg, Tl, Pb, Bi, Po, At, Rn
- Group 7 (VIIB):** Mn, Tc, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe
- Group 8 (VIII):** Fe, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, Te, I, Xe
- Group 9 (VIII):** Co, Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr
- Group 10 (VIII):** Ni, Cu, Zn, Ga, Ge, As, Se, Br, Kr
- Group 11 (IB):** Cu, Ag, Au
- Group 12 (IIB):** Zn, Cd, Hg
- Group 13 (IIIA):** B, Al, Ga, In, Tl, Pb, Bi, Po, At, Rn
- Group 14 (IVA):** C, Si, Ge, Sn, Pb, Bi, Po, At, Rn
- Group 15 (VA):** N, P, As, Sb, Te, I, Xe
- Group 16 (VIA):** O, S, Se, Te, I, Xe
- Group 17 (VIIA):** F, Cl, Br, I, Xe
- Group 18 (VIIIA):** He, Ne, Ar, Kr, Xe, Rn

Джон Дальтон (1800)